

■ Day 2 — Loading Data into Redshift using S3 + COPY Command

■ Topics Covered Today

- What is COPY Command
 - Amazon S3 + Redshift Integration
 - IAM Role for Redshift
 - Loading CSV Data into Redshift
 - Querying Large Data
 - Real Data Warehouse Flow
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■■ Real Industry Architecture

Applications / Databases

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Amazon S3

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Redshift COPY

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Amazon Redshift

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BI Dashboards

■ Step 1 — Create S3 Bucket

- Open Amazon S3 from AWS Console
 - Create bucket: **redshift-demo-bucket**
 - Keep Block Public Access ON
 - Upload sales_data.csv
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■ CSV Sample Data

```
order_id,customer_name,product,amount,city
1,Rohan,Laptop,75000,Mumbai
2,Amit,Phone,30000,Pune
3,Neha,Tablet,25000,Delhi
```

■ Step 2 — Create IAM Role

- Open IAM → Roles
- Create Role for Redshift
- Attach Permission: **AmazonS3ReadOnlyAccess**
- Role Name: **RedshiftS3Role**

■ Step 3 — Attach IAM Role to Namespace

- Open Redshift Serverless
 - Go to Namespace
 - Open Security & Encryption
 - Attach IAM Role
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■ Step 4 — Open Query Editor v2

- Open Query Editor v2
 - Connect using username & password
 - Select created workgroup
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■■ Step 5 — Create Table

```
CREATE TABLE sales (  
  order_id INT,  
  customer_name VARCHAR(100),  
  product VARCHAR(100),  
  amount INT,  
  city VARCHAR(100)  
);
```

■ Step 6 — COPY Command

```
COPY sales  
FROM 's3://redshift-demo-bucket/sales_data.csv'  
IAM_ROLE 'YOUR-IAM-ROLE-ARN'  
CSV  
IGNOREHEADER 1;
```

■ Analytics Queries

```
SELECT SUM(amount) FROM sales;
```

```
SELECT city, SUM(amount)  
FROM sales  
GROUP BY city;
```

■ Key Concepts Learned

- COPY command is optimized for large-scale loading
 - Redshift uses parallel processing
 - S3 acts as data lake storage
 - Redshift performs analytics very fast
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■ Day 2 Outcome

- Integrated S3 with Redshift
 - Created IAM Role
 - Loaded CSV data into Redshift
 - Executed analytics queries
 - Understood real-world pipeline
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